water or lemon extract it will still be a mixture. A mixture need not have a fixed proportion of components.

- A mixture is an impure substance and contains more than one kind of particles.
- In the mixture the components are mixed in any proportion.

A mixture can be a physical combination of two or more elements. Example: 22 carat gold which is composed of gold and copper or gold and cadmium.

It can be a physical combination of two or more compounds. Example: Aerated drink which is composed of carbon dioxide, water, sweetening and colouring agents.

It can be a physical combination of an element and a compound. Example: Tincture of iodine is composed of Iodine in alcohol.

3.6 Separation of Mixtures

Are all mixtures used as they are? Or is there a need for separating the components? Materials we use in our day-to-day life are got from different sources and are very often combined with other substances.

Mixtures like coffee and ice cream are taken as such. There is no need for separation of this substances. Metals occur in the form of ores under the earth's crust. But if we want to use a pure metal, we need to adopt a laborious process of extraction to separate the useful metal from the ore.

What is meant by separation? The process by which the components of mixture are isolated and removed from each other to get pure substance is called separation. To know about the original properties and uses of the individual substance we need separation.

When and why do we need to separate mixtures?

- ❖ When we need to remove impurities or harmful components from the mixtures. Eg. Stones from rice.
- ❖ When the useful component has to be separated from other components. Eg. Petrol from petroleum.
- ❖ When a substance has to be obtained in highly pure form. Eg. Gold from gold mines.

Let us visit Selvi's Family

It is 7 am and Selvi's family is busy. At home, in the kitchen, Selvi's mother is making tea for the family and her grandmother is separating butter from curds. Her father and uncle are out in the field collecting paddy after harvesting. Selvi is helping her mother to cook rice and is separating stones from the rice. Selvi's little brother Balu is fascinated by a piece of magnet that was given by his friend and is playing outside in the sand with it.

Can you list out in your note book, the different activities that Selvi's family members are engaged in?

Let us explore the different separating methods involved in the above activities and also learn about a few other methods.

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